

Lecture 12

The CKKS Scheme: Approximate Arithmetic

SEC 595: Encrypted Computing

Roadmap

BFV (Lecture 10) and BGV (Lecture 11) keep the RLWE noise below a threshold so that decryption is exact. CKKS (Cheon, Kim, Kim and Song, 2017) instead treats the noise as fixed-point rounding error and keeps it. Section 1 states the idea. Section 2 encodes complex vectors through the canonical embedding. Sections 3 and 4 give the scheme: keys, encryption, decryption, addition, multiplication and relinearization. Section 5 introduces RESCALE, which undoes the squaring of the scale, and Section 6 builds the modulus chain on it. Section 7 measures precision and Section 8 security. Section 9 runs the scheme end to end on a toy instance, and Section 10 places the three schemes side by side.

1. Noise as Rounding Error

1.1 The Idea

In every RLWE scheme decryption produces $c_0 + c_1 \text{SK} = (\text{scaled message}) + v$ with v small and unknown. BFV removes v by rounding, BGV by reduction modulo t . CKKS observes that a scaled message plus a small perturbation is exactly what fixed-point arithmetic stores anyway, and declares the noise part of the answer. The price is that correctness holds only approximately:

$$\text{Decrypt}(\text{Encrypt}(\mu)) \approx \mu, \quad \text{Decrypt}(\text{ct}_1 * \text{ct}_2) \approx \text{Decrypt}(\text{ct}_1) * \text{Decrypt}(\text{ct}_2).$$

The ring, the keys, relinearization and the modulus chain are inherited almost unchanged.

1.2 Fixed Point, Recalled

In Lecture 1 a real x is stored as $\hat{x} = \lfloor \Delta x \rfloor$ with $\Delta = 2^f$. Addition keeps the scale, $\hat{x} + \hat{y} \approx \Delta(x + y)$; multiplication squares it, $\hat{x}\hat{y} \approx \Delta^2 xy$, and the scale is restored by **rescaling**: dividing by Δ and rounding. Two facts carry the whole design.

Fact A. A large Δ makes a small error harmless: with error of size 2^k at scale 2^f , about $f - k$ bits of the value survive.

Fact B. Multiplication squares the scale; something must undo it.

Example 1 (The $Q_{4.4}$ round trip). Take $\Delta = 16$, $x = 2.75$, $y = 3.25$.

- (a) $\hat{x} = 16 \cdot 2.75 = 44$ and $\hat{y} = 16 \cdot 3.25 = 52$, both exact.
- (b) $\hat{x}\hat{y} = 2288$, at scale $\Delta^2 = 256$.
- (c) $\lfloor 2288/16 \rfloor = 143$ at scale 16, and $143/16 = 8.9375 = xy$.

- (d) With errors $e_x = 3$, $e_y = -2$: $(44 + 3)(52 - 2) = 2288 + 3 \cdot 52 - 2 \cdot 44 - 6 = 2350$, and $\lfloor 2350/16 \rfloor = \lfloor 146.875 \rfloor = 147$, i.e. 9.1875. The error is $0.25 = 2^{-2}$: two of the four fractional bits survive. Note that the terms linear in the errors (156 and -88) dominate the quadratic term (-6).

2. Encoding and Decoding

2.1 The Canonical Embedding and Slots

Let $K = \mathbb{Q}[X]/(X^n + 1)$ and $\zeta = \exp(\pi i/n)$, a primitive $2n$ -th root of unity; the roots of $X^n + 1$ are ζ^j for odd j (Lecture 4). Real coefficients give $a(\bar{w}) = \overline{a(w)}$, so the roots come in conjugate pairs and one root from each pair suffices.

Definition 1 (Canonical embedding).

$$\tau : K \rightarrow \mathbb{C}^{n/2}, \quad \tau(a) = (a(\zeta), a(\zeta^5), a(\zeta^{25}), \dots).$$

τ is an $\mathbb{R}$ -linear bijection (both sides have real dimension n) and is multiplicative, since evaluation at a point is a ring homomorphism: ring multiplication becomes slotwise multiplication. The $n/2$ coordinates are the **slots**; at $n = 2^{13}$ one ciphertext holds 4096 real or complex values. The representatives $\langle 5 \rangle$ are chosen so that $X \mapsto X^5$ rotates the slots cyclically.

For $n = 4$, τ^{-1} is explicit: with $(z_1, z_2) = \tau(a)$ and the conjugates filling the other two roots,

$$a_k = \frac{1}{4} \sum_{j \in \{1, 3, 5, 7\}} a(\zeta^j) \zeta^{-jk} = \frac{1}{2} \operatorname{Re}(z_1 \zeta^{-k} + z_2 \zeta^{-5k}),$$

because the evaluation matrix $A_{k,j} = \zeta^{kj}$ satisfies $A^* A = 4I$.

2.2 ENCODE and DECODE

$\tau^{-1}(z)$ has irrational coefficients in general. Fixed point repairs this.

Definition 2 (ENCODE, DECODE). For a scale factor $\Delta > 0$,

$$\text{ENCODE}(z, \Delta) = \lfloor \Delta \cdot \tau^{-1}(z) \rfloor \in R, \quad \text{DECODE}(a, \Delta) = \tau\left(\frac{a}{\Delta}\right) \in \mathbb{C}^{n/2},$$

with rounding applied coefficientwise.

These are Lecture 1's $\lfloor a \cdot 2^f \rfloor$ and $a/2^f$ after a change of basis that makes multiplication slotwise. Rounding moves each coefficient by at most $1/2$, so each slot moves by at most $n/2$ before and $n/(2\Delta)$ after the division by Δ . Δ wants to be large for precision; the modulus budget (Section 6) wants it small. That tension is the design of the scheme.

Example 2 (Song's toy encoding). Take $n = 4$, $\Delta = 10^2$, $z = (1+4i, 5-2i)$. With $\zeta^{-1} = (1-i)/\sqrt{2}$ and $\zeta^{-5} = \zeta^3 = (-1+i)/\sqrt{2}$,

$$a_1 = \frac{1}{2} \operatorname{Re}\left(\frac{(1+4i)(1-i) + (5-2i)(-1+i)}{\sqrt{2}}\right) = \frac{1}{2} \operatorname{Re}\left(\frac{(5+3i) + (-3+7i)}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}},$$

and similarly $a_0 = 3$, $a_2 = 1$, $a_3 = 5/\sqrt{2}$. Hence

$$\Delta \tau^{-1}(z) = (300, 50\sqrt{2}, 100, 250\sqrt{2}) = (300, 70.711, 100, 353.553).$$

Only the two irrational entries are rounded (by 0.289 and 0.447), giving

$$\mu(X) = 300 + 71X + 100X^2 + 354X^3.$$

Decoding, $\text{DECODE}(\mu, \Delta) = (0.99889 + 4.00520i, 5.00111 - 2.00520i)$, a slot error of 5.3×10^{-3} , under $n/(2\Delta) = 0.02$. At $\Delta = 10^3$ the same z gives $\mu = 3000 + 707X + 1000X^2 + 3536X^3$ and a slot error of 4.8×10^{-4} : ten times the scale, about a tenth of the error.

A second instance: $\Delta = 2^7$, $z = (1.2 - 3.4i, 5.6 + 7.8i)$. Then

$$\Delta\tau^{-1}(z) = (435.2, -705.9754, 281.6, -307.7329), \quad \mu = 435 - 706X + 282X^2 - 308X^3,$$

with

$$\mu(\zeta) = 2^7(1.19978 - 3.39849i), \quad \mu(\zeta^5) = 2^7(5.59710 + 7.80474i),$$

slot errors 1.5×10^{-3} and 5.6×10^{-3} .

3. Keys, Encryption, Decryption

3.1 KeyGen and Encrypt

Plaintexts are $\mathbb{C}^{n/2} \xrightarrow{\text{ENCODE}} R = \mathbb{Z}[X]/(X^n + 1)$ and ciphertexts lie in $R_{q_\ell}^2$ at a level ℓ (Section 6). There is **no plaintext modulus** t . The distributions are those of Lecture 6, §1: R_2 ternary, $\mathcal{X}$ a discrete Gaussian with bound β .

Definition 3 (KeyGen, Encrypt). Sample $\text{SK} \leftarrow R_2$, $a \leftarrow R_{q_L}$, $e \leftarrow \mathcal{X}$ and set

$$\text{PK} = ([-(a\text{SK} + e)]_{q_L}, a), \quad \text{so} \quad \text{PK}_0 + \text{PK}_1\text{SK} = -e.$$

To encrypt $\mu \in R$, sample $u \leftarrow R_2$, $e_1, e_2 \leftarrow \mathcal{X}$ and output

$$c_0 = [\text{PK}_0 u + e_1 + \mu]_{q_\ell}, \quad c_1 = [\text{PK}_1 u + e_2]_{q_\ell}.$$

Compare BFV's $c_0 = \text{PK}_0 u + e_1 + \Delta M$ and BGV's $c_0 = \text{PK}_0 u + t e_1 + M$. The only change from BFV is that the message carries no Δ : ENCODE has already applied it.

3.2 Decrypt

Definition 4 (Decrypt). Compute $\hat{\mu} = [c_0 + c_1\text{SK}]_{q_\ell}$, then output $\text{DECODE}(\hat{\mu}, \Delta)$.

There is no rounding, no division and no reduction modulo t : decryption is the bare evaluation $\text{ct}(\text{SK})$.

Proposition 1 (Decryption identity). *For a fresh ciphertext, $c_0 + c_1\text{SK} = \mu + v \pmod{q_\ell}$ with $v = -eu + e_1 + e_2\text{SK}$.*

Proof. $c_0 + c_1\text{SK} = \mu + e_1 + e_2\text{SK} + u(\text{PK}_0 + \text{PK}_1\text{SK}) = \mu + e_1 + e_2\text{SK} - eu$. □

The mask $au\text{SK}$ cancels as in BFV, and v is BFV's noise, so $\|v\| \leq n\beta + \beta + n\beta = 2n\beta + \beta$ (Lecture 10). Nothing removes it.

Proposition 2 (Approximate correctness). *If $\|\mu + v\| < q_\ell/2$, decryption returns $\mu + v$ exactly in R , and*

$$\text{DECODE}(\hat{\mu}, \Delta) = z + \frac{\tau(v)}{\Delta} + \frac{\tau(\varepsilon)}{\Delta},$$

where $\varepsilon = \Delta\tau^{-1}(z) - \mu$ is the encoding error. Otherwise the output is unrelated to z .

Proof. Over $\mathbb{Z}$, $c_0 + c_1\text{SK} = \mu + v + q_\ell k$. If $\|\mu + v\| < q_\ell/2$ the centred reduction removes $q_\ell k$ exactly; then $\mu = \Delta\tau^{-1}(z) - \varepsilon$ and $\mathbb{R}$ -linearity of τ give the formula. If not, a nonzero multiple of q_ℓ remains, which is enormous after division by Δ . $\square$

Correctness is only about wraparound: there is no threshold below which the answer becomes exact. Since $\mu \approx \Delta\tau^{-1}(z)$, the condition bounds the data: roughly $\|z\| < q_\ell/(2\Delta)$. At $q_0 = 2^{60}$, $\Delta = 2^{40}$ this allows values up to about 2^{20} , and overflow is silent.

4. EvalAdd, EvalMult, Relinearization

4.1 EvalAdd

Definition 5 (EvalAdd). For ciphertexts at the same level ℓ and the same scale Δ ,

$$\text{EvalAdd}(\text{ct}^{(1)}, \text{ct}^{(2)}) = ([c_0^{(1)} + c_0^{(2)}]_{q_\ell}, [c_1^{(1)} + c_1^{(2)}]_{q_\ell}).$$

The sum decrypts to $\mu_1 + \mu_2 + (v_1 + v_2)$, so $\|v_{\text{add}}\| \leq \|v_1\| + \|v_2\|$, the scale is unchanged and no level is consumed. The new condition is scale alignment: $\Delta z + \Delta' z'$ decodes to nothing meaningful unless $\Delta = \Delta'$.

4.2 EvalMult and the Squared Scale

Definition 6 (EvalMult).

$$\text{EvalMult}(\text{ct}^{(1)}, \text{ct}^{(2)}) = (c_0^{(1)}c_0^{(2)}, c_0^{(1)}c_1^{(2)} + c_1^{(1)}c_0^{(2)}, c_1^{(1)}c_1^{(2)}) \bmod q_\ell.$$

This is the product in $R_{q_\ell}[S]$ (Lecture 4), identical in BFV, BGV and CKKS. What differs is the plaintext arithmetic. With scaled values $\Delta x + e_x$ and $\Delta y + e_y$,

$$(\Delta x + e_x)(\Delta y + e_y) = \underbrace{\Delta^2 xy}_{\text{payload}} + \Delta(xe_y + ye_x) + e_x e_y,$$

while in BGV $(te_1 + M_1)(te_2 + M_2) = t^2 e_1 e_2 + t(e_1 M_2 + e_2 M_1) + M_1 M_2$. The same three terms appear with the roles of payload and noise exchanged. The scale becomes Δ^2 . A plaintext–ciphertext product $(\mu' c_0, \mu' c_1)$ needs no relinearization but also squares the scale.

Proposition 3 (Multiplicative noise growth). *If $\text{ct}^{(i)}(\text{SK}) = \mu_i + v_i$, the product decrypts to $\mu_1 \mu_2 + v_\times$ with $v_\times = \mu_1 v_2 + \mu_2 v_1 + v_1 v_2$ and*

$$\|v_\times\| \leq n(\|\mu_1\|\|v_2\| + \|\mu_2\|\|v_1\|) + n\|v_1\|\|v_2\|.$$

Proof. Expand $(\mu_1 + v_1)(\mu_2 + v_2)$ and apply $\|ab\| \leq n\|a\|\|b\|$ (Lecture 3) to each term. $\square$

In BGV the dangerous term is $t^2 e_1 e_2$, quadratic in the noise. In CKKS $\mu_i \approx \Delta z_i$ is far larger than v_i , so the *linear* terms dominate. Relative to Δ^2 , a multiplication costs a bit or two of precision and does not compound.

4.3 Relinearization

Definition 7 (EK and Relinearize). Publish $EK = ([-(aSK + e) + SK^2]_{q_\ell}, a)$, so $EK_0 + EK_1SK = SK^2 - e$. For $c^* = (c_0^*, c_1^*, c_2^*)$ set

$$\hat{c}_0 = c_0^* + EK_0 c_2^*, \quad \hat{c}_1 = c_1^* + EK_1 c_2^*.$$

Then $\hat{c}_0 + \hat{c}_1 SK = c_0^* + c_1^* SK + c_2^* SK^2 - c_2^* e$. This is BFV's construction unchanged. BGV needed a factor t in EK to protect the residue modulo t ; CKKS has no residue to protect, so $-c_2^* e$ is only more noise. With gadget base w and $\ell_w = \lceil \log_w q_\ell \rceil$ (Lecture 4) the added noise is

$$\|v_{\text{Relinearize}}\| \leq n \ell_w (w - 1) \beta.$$

5. Rescale

5.1 The Operation

Definition 8 (RESCALE). For $ct \in R_{q_\ell}^2$ at scale Δ^2 ,

$$\text{RESCALE}(ct, \Delta) = \lfloor \Delta^{-1} \cdot ct \rfloor \in R_{q_{\ell-1}}^2, \quad q_{\ell-1} = q_\ell / \Delta,$$

coefficientwise; the result has scale Δ .

Proposition 4. If $ct(SK) = \Delta^2 \mu + v \pmod{q_\ell}$, then $ct'(SK) = \Delta \mu + \Delta^{-1} v + r \pmod{q_{\ell-1}}$ with $\|r\| \leq \frac{1}{2}(1 + n\|SK\|)$.

Proof. Write $c'_i = \Delta^{-1} c_i + r_i$ with $\|r_i\| \leq \frac{1}{2}$. Over $\mathbb{Z}$, $c_0 + c_1 SK = \Delta^2 \mu + v + q_\ell k$, so $c'_0 + c'_1 SK = \Delta \mu + \Delta^{-1} v + q_{\ell-1} k + (r_0 + r_1 SK)$, and $\|r_0 + r_1 SK\| \leq \frac{1}{2} + \frac{1}{2} n\|SK\|$. $\square$

The payload is the same at scale Δ , the noise is divided by Δ , and rounding adds at most $\frac{1}{2}(1 + n\|SK\|)$.

5.2 MODSWITCH Under Another Name

	BGV MODSWITCH	CKKS RESCALE
Formula	$\lfloor (q'/q) ct \rfloor$	$\lfloor (q'/q) ct \rfloor$
Rounded	within the class mod t	to the nearest integer
Purpose	shrink the noise	restore $\Delta^2 \rightarrow \Delta$
Constraint	$q \equiv q' \pmod{t}$	none
Message	exactly preserved	divided by q/q'

BGV needs the congruence because it reduces the plaintext modulo t , and the term $(q - q')k$ would disturb the residue (Lecture 11). CKKS never reduces the plaintext modulo anything, so no congruence is needed and each rounding correction is at most $1/2$. Same formula, different disease: in BGV the *noise* explodes; in CKKS the *scale* does.

5.3 What RESCALE Does, and Does Not, Do

It does not improve precision: payload and noise are divided alike, so $\|v\|/\text{scale}$ is unchanged up to the rounding term. What it buys is the growth rate of the modulus. Without RESCALE, d multiplications leave scale Δ^{2^d} , so $\log q > 2^d \log \Delta$. With it, the scale returns to Δ each time, so $\log q > L \log \Delta$. Exponential becomes linear: that is Fact B.

Example 3 (A rescale trace). At level 1, $q_1 = 100000190000$, a ciphertext at scale $\Delta^2 = 10^8$ is (this is the relinearized product of Section 9)

$$\begin{aligned}\hat{c}_0 &= -6564956194 + 25551957201X + 21920360407X^2 + 28669366452X^3, \\ \hat{c}_1 &= -9461472835 + 40573592885X - 25296998807X^2 - 43550170004X^3,\end{aligned}$$

with $\Delta = 10^4$, $SK = 1 - X + X^3$ and noise $\|v\| = 117420$. Dividing by Δ and rounding, with corrections $c' - \Delta^{-1}c$:

	$\Delta^{-1}\hat{c}_0$	correction	$\Delta^{-1}\hat{c}_1$	correction
1	-656495.6194	-0.3806	-946147.2835	+0.2835
X	2555195.7201	+0.2799	4057359.2885	-0.2885
X^2	2192036.0407	-0.0407	-2529699.8807	-0.1193
X^3	2866936.6452	+0.3548	-4355017.0004	+0.0004

Every correction is at most $1/2$ in size. The new modulus is $q_0 = q_1/\Delta = 10000019$ and

$$\begin{aligned}c'_0 &= -656496 + 2555196X + 2192036X^2 + 2866937X^3, \\ c'_1 &= -946147 + 4057359X - 2529700X^2 - 4355017X^3.\end{aligned}$$

Evaluating at SK modulo q_0 gives $-15000 + 88383X - 40006X^2 + 95473X^3$. Against the rounded payload $\lfloor \mu^{(1)}\mu^{(2)}/\Delta \rfloor = -15001 + 88388X - 40001X^2 + 95460X^3$ (the two ties 88388.5 and 95460.5 rounded down), the noise is

$$v' = 1 - 5X - 5X^2 + 13X^3, \quad \|v'\| = 13.$$

Before the switch $\|v\|/\Delta = 117420/10^4 = 11.74$; the extra 1.26 is rounding, within the bound $\frac{1}{2}(1 + n\|SK\|) = \frac{1}{2}(1 + 4) = 2.5$ of Proposition 4.

6. Levels

6.1 The Modulus Chain

Definition 9 (Modulus chain). Fix a base modulus $q_0 \gg \Delta$ and a depth L , and set $q_\ell = q_0 \cdot \Delta^\ell$ for $0 \leq \ell \leq L$, $q = q_L$. A ciphertext with modulus q_ℓ is at **level** ℓ .

Operation	in	out	costs
EvalAdd	(ℓ, Δ)	(ℓ, Δ)	nothing
EvalMult	(ℓ, Δ)	(ℓ, Δ^2)	a third component
Relinearize	(ℓ, Δ^2)	(ℓ, Δ^2)	a little noise
RESCALE	(ℓ, Δ^2)	$(\ell - 1, \Delta)$	one level

Level is computational capability. At level 0 only addition and decryption remain: there is no room for a squared scale. q_0 is chosen $\gg \Delta$ so that the final payload still fits (Proposition 2).

6.2 The Depth Budget

Taking logarithms of the chain,

$$\log q > L \cdot \log \Delta,$$

with slack $\log q_0$. For fixed security a larger $\log q$ forces a larger n (Lecture 17), and every operation becomes slower: depth, precision and speed are one resource. Circuit shape matters: $(xy)(zw)$ is a balanced tree of depth 2 and needs $q_0\Delta^2$, while $((xy)z)w$ is a chain of depth 3 and needs $q_0\Delta^3$, for the same three multiplications. Compiling for CKKS means minimising depth, not operation count.

6.3 Song's Parameter Set

Ring dimension	$n = 2^{13}$
Base modulus	$\log_2 q_0 = 60$
Scale	$\log_2 \Delta = 40 = \log_2 q_1 = \log_2 q_2$
Special prime	$\log_2 q_3 = 60$
Total	$60 + 40 + 40 + 60 = 200 \leq 218$ (security ceiling)
Depth	$L = 2$
Precision	roughly 30 bits
Data range at level 0	$q_0/\Delta = 2^{20}$

q_3 is not a level: the keys live at $q_0 q_1 q_2 q_3$ so that key-switching noise can be divided away. In practice $q_\ell = q_0 q_1 \cdots q_\ell$ in RNS (Lecture 4) with primes $q_i \approx \Delta$. RESCALE then divides by q_ℓ , not Δ , so the scale drifts: $\Delta' = \Delta^2/q_\ell \approx \Delta$, and libraries track the scale per ciphertext.

7. Precision

Proposition 5 (Precision). *A ciphertext at scale Δ with noise v decodes to slots correct to about $\log_2 \Delta - \log_2 \|v\|$ bits.*

Sketch. By Proposition 2 the slot error is $\tau(v)/\Delta$ plus the smaller encoding term, of size about $\|v\|/\Delta$; take $-\log_2$. $\square$

Song reports $\log_2 \|v\| \approx 10$ –15 for basic operations, so $\log_2 \Delta = 40$ gives 25–30 bits. The error analysis of a CKKS circuit is the error analysis of the same circuit in floating point.

Example 4 (Precision calculations). (a) At $\Delta = 2^{40}$ and $\|v\| = 2^{12}$: $40 - 12 = 28$ bits, i.e. $28 \log_{10} 2 \approx 8.4$ decimal digits.
(b) At $\Delta = 10^4$ with $\|v'\| = 13$: predicted slot error $13/10^4 = 1.3 \times 10^{-3}$.
(c) After one multiplication, $\|v_\times\| \approx n\|\mu\|\|v\|$ with $\|\mu\| \approx \Delta\|z\|$, at scale Δ^2 , so

$$\log_2 \frac{\Delta^2}{n\|\mu\|\|v\|} \approx \log_2 \frac{\Delta}{n\|z\|\|v\|} \text{ bits.}$$

(d) With $\|v\| \approx 2^{10}$ and $n\|z\| \approx 2^{16}$, 30 bits require $\log_2 \Delta = 30 + 16 + 10 = 56$. In practice one takes $\Delta = 2^{40}$ and accepts precision in the twenties.

8. Security

8.1 IND-CPA and IND-CPA⁺

PK is an RLWE sample, so IND-CPA (Lecture 8, §1.5) follows from decision RLWE (Lecture 6, §2.2) exactly as for BFV and BGV. A stronger model captures a client that decrypts and publishes the result.

Definition 10 (IND-CPA⁺, Li–Micciancio 2021). The adversary may adaptively (1) request encryptions of message pairs (m_0, m_1) , receiving $\text{Encrypt}(m_b)$; (2) request homomorphic evaluation of any circuit on its ciphertexts; (3) request decryption of an evaluated ciphertext, provided the result is the same under $b = 0$ and $b = 1$. The scheme is secure if $|\Pr[b' = b] - \frac{1}{2}|$ is negligible.

For an exactly correct scheme (BFV, BGV) the decryption oracle returns a value the adversary already knows, so it can be simulated and IND-CPA^+ coincides with IND-CPA .

8.2 The Li–Micciancio Attack

In CKKS a decryption query on (c_0, c_1) returns $\hat{\mu} = [c_0 + c_1 \text{SK}]_q = \mu + v$, the payload *plus the noise*. Hence

$$\text{SK} = (\hat{\mu} - c_0) \cdot c_1^{-1} \quad \text{in } R_q,$$

computed by the extended Euclidean algorithm (Lecture 1); a random c_1 is invertible with high probability. In BFV and BGV the oracle returns M , and v remains unknown. One published decryption recovered the key against every major CKKS library.

Example 5 (Key recovery in miniature). Work in $R_{17} = \mathbb{Z}_{17}[X]/(X^2 + 1)$. Since $4^2 = 16 \equiv -1$, $X^2 + 1 = (X - 4)(X + 4)$ and the ring is not a field. The adversary sees $c_0 = 6 + 6X$, $c_1 = 2 + X$, $\hat{\mu} = 7 + 2X$.

- (a) $c_1(4) = 6$ and $c_1(-4) = -2 \equiv 15$ are nonzero, so c_1 is invertible. Writing $(2 + X)(a + bX) = (2a - b) + (a + 2b)X = 1$ gives $a = -2b$ and $-5b = 1$, so $b = -5^{-1} = -7 \equiv 10$ and $a = -20 \equiv 14$: $c_1^{-1} = 14 + 10X$. Check: $(2 + X)(14 + 10X) = 18 + 34X \equiv 1$.
- (b) $\hat{\mu} - c_0 = 1 - 4X \equiv 1 + 13X$, and $\text{SK} = (1 + 13X)(14 + 10X) = (14 - 130) + (10 + 182)X = -116 + 192X \equiv 3 + 5X$.
- (c) Check: $c_1 \text{SK} = (2 + X)(3 + 5X) = 1 + 13X$, and $c_0 + c_1 \text{SK} = 7 + 19X \equiv 7 + 2X = \hat{\mu}$.

8.3 The Fix: Noise Flooding

Publish $\hat{\mu} + e_{\text{flood}}$ with fresh e_{flood} large enough that the output is statistically independent of v . A statistical distance of $2^{-\lambda}$ needs $\|e_{\text{flood}}\| \approx 2^\lambda \|v\|$, costing about λ bits of modulus and of precision. Every current library implements a variant. A client that decrypts locally and keeps the answer is unaffected; a client that returns the decryption to the server is completely broken without it.

9. A Verified Instance

9.1 Setup and Keys

Take $n = 4$, $\Delta = 10^4$, $L = 1$, $q_0 = 10000019$ (prime) and $q_1 = q_0 \Delta = 100000190000$. Small polynomials are ternary, so $\beta = 1$ and the fresh bound is $2n\beta + \beta = 9$. Let $\text{SK} = 1 - X + X^3$, $e = 1 - X^2 + X^3$, and

	1	X	X^2	X^3
$a = \text{PK}_1$	48772087225	71480653741	58975695645	69681953716
PK_0	-46973387201	36267129129	-17813278187	40521844703

Then $\text{PK}_0 + \text{PK}_1 \text{SK} = -1 + X^2 - X^3 = -e$. The level-0 data range is $q_0/\Delta \approx 10^3$ and the wraparound budget is $q_0/2 = 5000009.5$.

9.2 Encoding and Encryption

Let $z^{(1)} = (1 + 4i, 5 - 2i)$ and $z^{(2)} = (2 + i, 1 - 3i)$. As in Example 2,

$$\begin{aligned}\Delta\tau^{-1}(z^{(1)}) &= (30000, 7071.068, 10000, 35355.339), \\ \mu^{(1)} &= 30000 + 7071X + 10000X^2 + 35355X^3, \\ \Delta\tau^{-1}(z^{(2)}) &= (15000, 17677.670, -10000, 10606.602), \\ \mu^{(2)} &= 15000 + 17678X - 10000X^2 + 10607X^3.\end{aligned}$$

The round-trip slot errors are 3.5×10^{-5} and 5.2×10^{-5} , under $n/(2\Delta) = 2 \times 10^{-4}$.

Encrypt with $u^{(1)} = X$, $e_1^{(1)} = -1 + X^2 + X^3$, $e_2^{(1)} = 1 + X$, and $u^{(2)} = -1 - X + X^2$, $e_1^{(2)} = X^2 + X^3$, $e_2^{(2)} = 1 - X + X^2$:

	1	X	X^2	X^3
$c_0^{(1)}$	-40521814704	-46973380130	36267139130	-17813242831
$c_1^{(1)}$	30318236285	48772087226	-28519536259	-41024494355
$c_0^{(2)}$	5308335091	-29815568953	34572941858	13558573221
$c_1^{(2)}$	-38065829153	10065685317	18315927840	42823194380

Computing $[c_0 + c_1\text{SK}]_{q_1} - \mu$, and independently $-eu + e_1 + e_2\text{SK}$, gives the same noises

$$v_1 = -X + 3X^3, \quad \|v_1\| = 3, \quad v_2 = 1 - X + X^2 + X^3, \quad \|v_2\| = 1,$$

both under 9. $\text{ct}^{(1)}$ decodes to $(0.999736 + 4.000113i, 5.000264 - 2.000113i)$, slot errors 2.9×10^{-4} ; the prediction of Proposition 5 is $\log_2 10^4 - \log_2 3 = 11.7$ bits, an error near $2^{-11.7} = 3.0 \times 10^{-4}$.

9.3 Addition and Multiplication

Add. The noise is $v_1 + v_2 = 1 - 2X + X^2 + 4X^3$, of norm $4 = 3 + 1$. The sum decodes to $(2.99969 + 5.000264i, 6.00051 - 5.000064i)$; the true value is $(3 + 5i, 6 - 5i)$.

Multiply. The tensor product decrypts, under $(1, \text{SK}, \text{SK}^2)$ modulo q_1 , to $\mu^{(1)}\mu^{(2)} + v_\times$ with

$$\begin{aligned}\mu^{(1)}\mu^{(2)} &= -150007787 + 883885000X - 400009347X^2 + 954605000X^3, \\ v_\times &= 5861 - 53288X - 51927X^2 + 117428X^3, \quad \|v_\times\| = 117428.\end{aligned}$$

At scale $\Delta^2 = 10^8$ this decodes to $(-2.001292 + 8.999928i, -0.998746 - 17.001154i)$; the true product is $z^{(1)} \odot z^{(2)} = (-2 + 9i, -1 - 17i)$. The product is already correct, at the wrong scale.

Example 6 (Decomposing the multiplication noise). By Proposition 3, $v_\times = \mu^{(1)}v_2 + \mu^{(2)}v_1 + v_1v_2$. In $\mathbb{Z}[X]/(X^4 + 1)$:

$$\begin{aligned}\mu^{(1)}v_2 &= 48284 - 68284X - 2426X^2 + 62426X^3, & \|\cdot\| &= 68284, \\ \mu^{(2)}v_1 &= -42427 + 15000X - 49499X^2 + 55000X^3, & \|\cdot\| &= 55000, \\ v_1v_2 &= 4 - 4X - 2X^2 + 2X^3, & \|\cdot\| &= 4.\end{aligned}$$

The constant terms add to $48284 - 42427 + 4 = 5861$, and likewise for the other coefficients, recovering $v_\times$. The bound is $4(35355 \cdot 1 + 17678 \cdot 3) + 4 \cdot 3 \cdot 1 = 353568$, against the observed 117428. The quadratic term contributes 4; the linear terms, proportional to the payload, contribute everything else. In BGV (Lecture 11) the payload M is small and the noise carries the factor t , so $t^2e_1e_2$ dominates; in CKKS the payload carries the large factor Δ , so the linear terms dominate.

9.4 Relinearize, Then Rescale

With $w = 2$, $\ell_w = \lceil \log_2 100000190000 \rceil = 37$. After Relinearize the noise is $5869 - 53272X - 51934X^2 + 117420X^3$, of norm 117420: no coefficient moves by more than 16, against the bound $n\ell_w(w-1)\beta = 4 \cdot 37 \cdot 1 \cdot 1 = 148$. Rescaling by Δ is Example 3, which gives $\|v'\| = 13$ in R_{q_0} with $\|\mu' + v'\| = 95473 < q_0/2$. The result decodes to

$$(-2.001339 + 8.999982i, -0.998661 - 17.001182i),$$

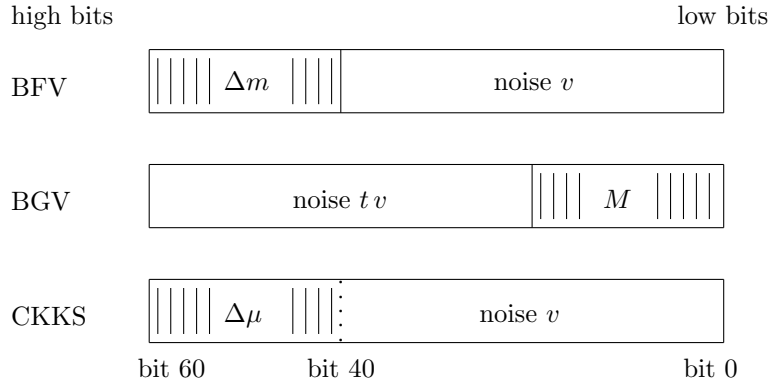
slot errors 1.34×10^{-3} and 1.79×10^{-3} . The prediction is $\log_2 10^4 - \log_2 13 = 13.29 - 3.70 = 9.59$ bits; observed $-\log_2(1.34 \times 10^{-3}) = 9.54$ and $-\log_2(1.79 \times 10^{-3}) = 9.13$.

Stage	modulus	scale	$\ v\ $
fresh	q_1	10^4	3, 1
after EvalAdd	q_1	10^4	4
after EvalMult	q_1	10^8	117428
after Relinearize	q_1	10^8	117420
after RESCALE	q_0	10^4	13

In the last step modulus, scale and noise all fall by a factor Δ : precision is unchanged, and the next multiplication starts from scale 10^4 , not 10^8 .

10. Where the Message Sits

Draw one ciphertext coefficient as a 60-bit word, with $t = 2^{20}$ and $\Delta = 2^{40}$; hatching marks the message.



BFV places the message in the top twenty bits and BGV in the bottom twenty; each draws a boundary inside the word and defends it. CKKS places $\Delta\mu$ at the top with no boundary below (dotted): payload and noise form one block and cannot be separated after encryption.

	BFV	BGV	CKKS
Plaintext	exact, in R_t	exact, in R_t	approximate, in $\mathbb{C}^{n/2}$
$c_0 + c_1\text{SK}$	$\Delta M + v$	$M + tv$	$\mu + v$
Δ applied at	encryption	—	<i>encoding</i>
Noise removed by	rounding	mod t	<i>not removed</i>
Correct when	$\ v\ < q/(2t)$	$\ v\ < q_\ell/(2t)$	never exactly; $\ \mu + v\ < q_\ell/2$
Level control	in EvalMult	MODSWITCH	RESCALE
Chain constraint	—	$p_i \equiv 1 \pmod{t}$	none

In CKKS the noise is never removed; it costs $\log_2 \|v\|$ bits of precision.

Summary

1. CKKS treats the RLWE noise as fixed-point rounding error: $\text{ENCODE}(z, \Delta) = \lfloor \Delta \tau^{-1}(z) \rfloor$ via the canonical embedding, with $n/2$ complex slots and encoding error at most $n/(2\Delta)$ per slot.
2. KeyGen and Encrypt are BFV's without Δ on the message and without t ; Decrypt is $[c_0 + c_1 \text{SK}]_{q_\ell} = \mu + v$, correct only up to the noise and only while $\|\mu + v\| < q_\ell/2$.
3. EvalMult squares the scale; its noise is dominated by the linear terms $\mu_1 v_2 + \mu_2 v_1$. Relinearize is BFV's.
4. $\text{RESCALE} = \lfloor \Delta^{-1} \text{ct} \rfloor$ drops one level of $q_\ell = q_0 \Delta^\ell$; it is BGV's MODSWITCH formula without the congruence. The budget is $\log q > L \log \Delta$, and precision is about $\log_2 \Delta - \log_2 \|v\|$ bits.
5. Vanilla CKKS fails IND-CPA⁺: one published decryption gives $\text{SK} = (\hat{\mu} - c_0) c_1^{-1}$. The fix is noise flooding.

Exercises

1. Repeat Example 1(d) in $Q_{4,12}$ ($\Delta = 2^{12}$) with the same errors $e_x = 3$, $e_y = -2$. How many fractional bits survive? State the rule relating this to f and to the size of the errors.
2. At $n = 4$, $\Delta = 10^2$, encode $z' = (2, 3)$ and multiply the result by the μ of Example 2 in $\mathbb{Z}[X]/(X^4 + 1)$. Decode at scale Δ^2 and check that you obtain approximately $(2(1 + 4i), 3(5 - 2i))$. Which property of τ makes this work?
3. In the instance of Section 9, encrypt the constant $\mu = 10000$ with $u = 1$, $e_1 = e_2 = 0$. Compute $[c_0 + c_1 \text{SK}]_{q_1}$ and the noise v two ways. Why is $v \neq 0$?
4. Take $c = -6564956194$ from Example 3. Round $\Delta^{-1}c$ instead to the nearest integer congruent to c modulo 8, as BGV with $t = 8$ would. How large is the correction now, and why does CKKS not need this?
5. Song evaluates $F(x) = \pi x^3 + 0.4x + 1$ on an encrypted x : square x and rescale; compute πx by a plaintext-ciphertext product and rescale; multiply the two and rescale; add the remaining terms. Starting from $(\text{level}, \text{scale}) = (2, \Delta)$, list the pair for every intermediate ciphertext. Where must two ciphertexts at different levels be aligned before adding?
6. For Song's parameter set, how many further 40-bit levels fit under the ceiling of 218 bits? A user needs 40 bits of precision; give two ways to obtain it and the cost of each.
7. In Example 5, suppose the oracle returned only $[\hat{\mu}]_4$, as an exact scheme with $t = 4$ would. Explain why the equation for SK no longer determines it.